University of Colombo, Sri Lanka

UCSC *University of Colombo School of Computing* BIT

DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY (EXTERNAL)

Academic Year 2021 — 2nd Year Examination — Semester 3

IT3206 — Data Structures and Algorithms

Part 1 - Multiple Choice Question Paper

(1 Hour)

Important Instructions

- This paper has **two (2) parts**, **Part 1** and **Part 2**.
- The duration of this part (Part 1) is **1 Hour**.
- The medium of instructions and questions is English.
- This paper has **25 questions** on **09 pages**. Answer **all** questions.
- All questions are of the **MCQ** (Multiple Choice Questions) type.
- Each question will have **5 (five)** choices with **one or more** correct answers.
- This paper consists of 100 marks and all the questions will carry equal marks.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Calculators are **not** allowed.
- *All Rights Reserved.* This question paper can NOT be used without proper permission from the University of Colombo School of Computing.

1) Consider the standard algorithm for determining whether a sequence of parentheses is balanced or not.

Suppose you run the algorithm on the following set of parentheses.

{ [ ] ([[ ]]) ([ ]) }

What is the maximum number of parentheses that will appear on the stack at a time during the above computation?

(a) 2	(b) 3	(c) 4
(d) 5	(e) 6	

2) Which of the following is/are **not** suited as an application of the queue data structure?

(a) Managing requests on a single shared resource such as CPU scheduling and disk
(b) Backtracking
(c) Data is transferred asynchronously
(d) Implementing a dictionary
(e) Reversing a set of characters in a word.

3) Five processes (P, Q, R, S and T) are waiting in a queue. Process R is waiting between Process P, and Process T. Process P is just behind Process Q, and Process Q is second in the queue.

Which of the processes is placed in the middle of the queue?

(a) P	(b) Q	(c) R
(d) S	(e) T	

4) Insertion sort is used to sort the following integers in ascending order.

34, 08, 64, 31, 32, 21, 09

What would be the array elements after the third pass of the sort?

(a) 08, 34, 64, 09, 32, 21, 31	(b) 08, 31, 34, 64, 32, 21, 09	(c) 08, 31, 32, 34, 64, 21, 09
(d) 08, 21, 31, 32, 34, 64, 09	(e) 08, 09, 21, 31, 32, 34, 64	

5) Which of the following is/are related to the worst-case scenario of the Bubble Sort?

(a) The Time complexity is $O(n^2)$
(b) The Time complexity is $O(n)$
(c) The array is stored in reverse order
(d) When the array is greater than hundred (100) elements
(e) When the array is less than ten (10) elements

6) Suppose that one sorts the following elements using the quicksort algorithm. Also, that they have selected the first element of the array or sub-array as the pivot value for the partitioning.

7, 67, 20, 16, 9, 4, 11, 75, 5, 89

Which of the acceptable pivot values can be chosen to fulfil the above requirement during the partitioning process?

(a) 5	(b) 7	(c) 20
(d) 67	(e) 89	

7) Consider the following data set.

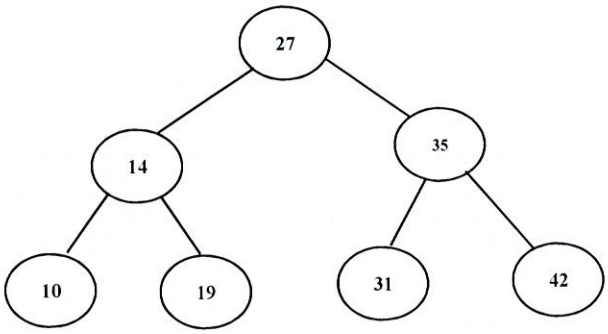
9, 21, 45, 56, 62, 70, 77, 81, 90, 100

How many comparisons are needed to find element number 70 using the iterative linear search algorithm?

.

(a) 3	(b) 4	(c) 5
(d) 6	(e) 7	

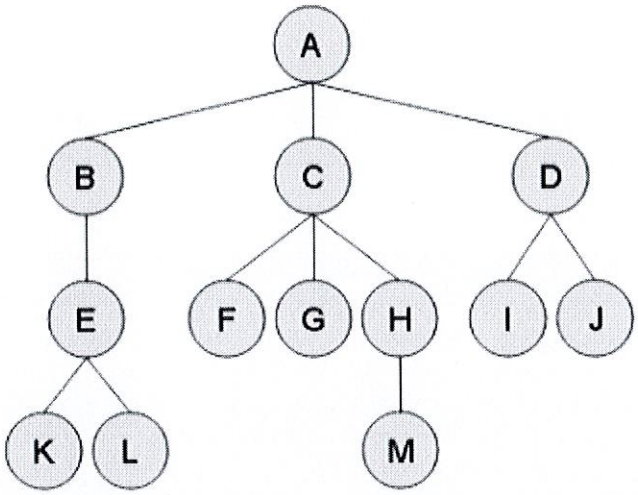
8) Consider the following tree



Which of the following statements is/are valid in connection to the above tree?

(a) The tree satisfies the Binary Tree properties, but it violates the Binary Search Tree Properties.
(b) If one inserts node number 5 as the left child of node number 10, it violates the AVL properties.
(c) Node numbers 10,19,31, and 42 are siblings of node 14.
(d) The height of the tree is 2.
(e) If one inserts node numbers 32 and 33 to the tree, it violates the AVL properties, and a single rotation is needed to restore the balance.

9) Consider the following tree.



Which of the following statements is/are valid in connection with the above tree?

(a) In the worst case, we need a maximum of 3 links for each node to implement the tree using a pointer-based implementation.
(b) In the best case, two pointers are needed to implement the tree, with the two pointers being to the left child and the right child.
(c) In the best case, two pointers are needed to implement the tree, with the two pointers being to the left most child and its right sibling.
(d) The breadth-first traversal order of the tree is a, b, c, d, e, f, g, h, i, j, k, and m.
(e) The proper ancestor of Node C is both Node C and Node A.

10) Which of the following are linear data structures?

(a) Stack	(b) Red-Black tree	(c) AVL tree
(d) Queue	(e) Circular array	

11) There exists a special type of a balanced binary tree where the root node is compared with its children and arranged accordingly. If node a has child b, then,

key(a) <= key(b)

Note that key(a) is the data portion of node a. If the value of parent is less than that of the child, this property generates a specific type of a tree data structure. Select the **most appropriate** data structure that will fit into the above scenario.

(a) Red-Black tree	(b) AVL tree	(c) Min heap
(d) Max heap	(e) Binary search tree	

12) You are given a graph where nodes are connected to each other with a number of edges. You wish to find connected components of the graph.

What is the best approach you will use to identify connected components in the graph?

(a) Breadth-first search	(b) Depth-first search	(c) Dijkstra algorithm
(d) Bellman Ford algorithm	(e) Prim's algorithm	

13) You are given an unweighted graph that needs to be stored in your computer. When you are storing, you should consider a faster lookup (i. e, O (1)) characteristic to save time on searching an element rather than the space efficiency characteristic.

Which representation would be the optimal choice for storing such a graph?

(a) Adjacency list	(b) Adjacency matrix	(c) Linked list
(d) Queue	(e) Doubly linked list	

14) There exists an array named `my_array` with length 10, as defined using JAVA syntax below.

```
int [] my_array = new int[10];
```

Which of the following statements is/are TRUE regarding `my_array`?

(a) my_array indexing begins with 0.
(b) my_array[-1] throws a run time error.
(c) my_array[10] returns the final data element of the array.
(d) my_array is an integer array with 10 integer elements.
(e) Indexes of my_array range from 0 to 8.

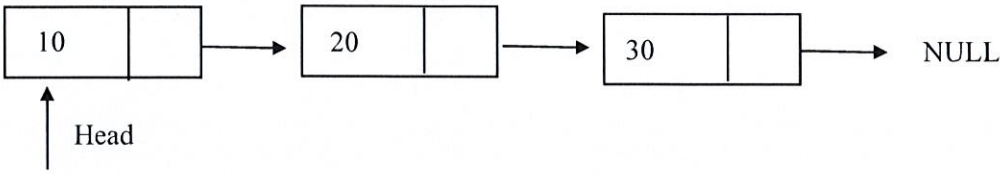
15) Suppose you need to store the following table as a single data structure.

	Mark 1	Mark 2
Student 1	45	55
Student 2	34	78

Which of the following statements is/are TRUE regarding the above table that needs to be stored as a single data structure?

(a) It is possible to create two single-dimensional arrays called student and marks.
(b) int []student = new int[2]; and int[] mark = new int[]; define two single 1-D arrays to store the above table.
(c) It is possible to create a single 2- dimensional arrays with 2 rows and 2 columns.
(d) int [][] array = new int [2][2] defines the 2-D array.
(e) array [row_number][column_number] can be used to access elements in the table for the 2-D array scenario.

16) Suppose you are given a single linked list as shown in the following representation.



Which of the code segments below correctly represents deletion of the head-node from the above singly linked list representation?

(a) Node temp = head;
head = head. next;

(b) Node temp = head;
head =next;

(c) Node temp;
temp = temp. next;

(d) Node temp = head;
head. next = head

(e) Node temp = head;
head. next = head

17) Which of the following statements is/are true?

(a) Recursive solutions have a better time complexity than non-recursive solutions.
(b) Recursive methods usually take more memory space than non-recursive methods.
(c) A recursive method can always be replaced by a non-recursive method.
(d) Recursive methods allow constructing clean, simple solutions to complex problems.
(e) Java uses Queue data structure to manage recursion.

18) Consider the following method used to reverse a string recursively. Note the blank indicated as ___(A)___.

```
public static String reverse(String string) {
    if (string.isEmpty()) {
        return string;
    } else {
        return ___(A)___
    }
}
```

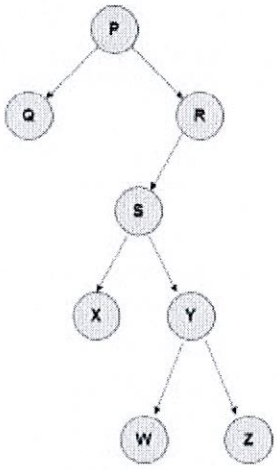
Which of the following would correctly fill the blank indicated by ___(A)___?

(a) string. charAt(0) + reverse(string. substring(1))
(b) reverse(string. substring(1)) + string. charAt(0);
(c) reverse(string. substring(1) + string. charAt(0));
(d) string. charAt(1) + reverse(string. substring(0));
(e) reverse(string. charAt(0) + string. substring(1));

19) What will be the worst-case time complexity for searching an item in a **Balanced Binary Search Tree**?

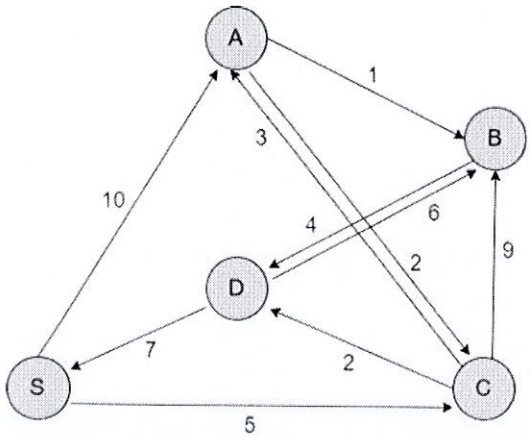
(a) $O(n^2)$	(b) $O(n)$	(c) $O(n \log n)$
(d) $O(\log n)$	(e) $O(\log n^2)$	

20) What is the correct pre-order, in-order and post-order traversals for the following tree?



(a) PQRSXYWZ, QPXSWYZR, QXWZYSRP	(b) PQSRXYWZ, QPSXWYZR, QXWZYSRP	(c) PQSRXYWZ, QPXSWYZR, QXWYZSRP
(d) PQRSXYWZ, QXPSWYZR, QXZWYSRP	(e) PQRSYXWZ, QPXSWZYR, QXWZYRSP	

21) What is/are the shortest paths from node S to node B?



(a) S → A → C → B	(b) S → C → A → B	(c) S → C → D → B
(d) S → C → B	(e) S → A → B	

22) Which of the following statements is/are true?

V – Number of Vertexes in the Graph
E – Number of Edges in the Graph

(a) Bellman-Ford algorithm can be used to find the shortest paths in graphs with negative weight cycles.
(b) The running time of the Bellman-Ford algorithm is O(VE)
(c) Interpolation is the basic principle behind the Bellman-Ford algorithm.
(d) The outer loop of the Bellman-Ford algorithm is executed V-1 times.
(e) Bellman-Ford algorithm is an example of Dynamic Programming.

23) What is the best-case time complexity for the Insertion Sort Algorithm?

(a) $O(n^2)$	(b) $O(n)$	(c) $O(n \log n)$
(d) $O(\log n)$	(e) $O(\log n^2)$	

24) What is the time complexity of the following code segment?

```
int p, q, r = 0;

for (p = n / 2; p < n; p++){
    for(q = 2; q < n; q = q){
        r += q;
        q = q * 2;
    }
}
```

(a) $O(n^2)$	(b) $O(n)$	(c) $O(n \log n)$
(d) $O(\log n)$	(e) $O(\log n^2)$	

25) There are n number of unsorted arrays that contain n number of elements each. Elements of all arrays are unique. What is the worst-case time complexity to calculate the median of medians of the n arrays?

(a) $O(n^2)$	(b) $O(n)$	(c) $O(n \log n)$
(d) $O(\log n)$	(e) $O(\log n^2)$	

******